

Problem 1. Let X and Y be two sets, and let $f : X \rightarrow Y$ be a function.

- (i) What does it mean to say that a function $Y \rightarrow X$ is *inverse* to f ?

Solution. We say $f^{-1} : Y \rightarrow X$ is an inverse to f if $f \circ f^{-1} = \mathbf{1}_Y$ and $f^{-1} \circ f = \mathbf{1}_X$.

- (ii) What does it mean to say that f is *bijective*?

Solution. We say f is bijective if it has an inverse.

(iii) Let Z be another set, and let $g : Y \rightarrow Z$ be another function. Suppose that f and g are bijective. Show directly from the definitions that $g \circ f$ is bijective.

Solution. If f^{-1} (resp. g^{-1}) is an inverse to f (resp. g), then $f^{-1} \circ g^{-1}$ is an inverse to $g \circ f$ because

$$\begin{aligned} f^{-1} \circ g^{-1} \circ g \circ f &= f^{-1} \circ \mathbf{1}_Y \circ f = f^{-1} \circ f = \mathbf{1}_X \\ g \circ f \circ f^{-1} \circ g^{-1} &= g \circ \mathbf{1}_Y \circ g^{-1} = g \circ g^{-1} = \mathbf{1}_Z. \end{aligned}$$

Problem 2.

- (i) Let $\sim$ be an equivalence relation on a set X , and write $\pi : X \rightarrow X/\sim$ for the quotient map. Let Y be another set, and let $F : X \rightarrow Y$ be a function. State the criterion on F for there to exist a function $f : X/\sim \rightarrow Y$ satisfying

$$f \circ \pi = F.$$

Solution. Such a function exists if and only if

$$x \sim x' \quad \Rightarrow \quad F(x) = F(x')$$

for all $x, x' \in X$.

- (ii) Let n be an integer. We defined the set of integers mod n to be $\mathbf{Z}/n\mathbf{Z} = \mathbf{Z}/\sim_n$ for a certain equivalence relation $\sim_n$ on $\mathbf{Z}$. Give the definition of $\sim_n$.

Solution. The definition is:

$$a \sim_n a' \quad \Leftrightarrow \quad n|(a - a').$$

(iii) Let m and n be integers. Use the criterion from (i) to construct a function

$$\mu : \mathbf{Z}/n\mathbf{Z} \rightarrow \mathbf{Z}/mn\mathbf{Z}$$

satisfying

$$\mu([a]_n) = [m \cdot a]_{mn} \quad \text{for all } a \in \mathbf{Z}.$$

Solution. We apply the criterion of (i) for $(X, \sim)$ being $(\mathbf{Z}, \sim_n)$ and F being the function

$$\mathbf{Z} \rightarrow \mathbf{Z}/mn\mathbf{Z} \quad a \mapsto [m \cdot a]_{mn}.$$

So we need to verify that the implication

$$a \sim_n a' \quad \Rightarrow \quad [m \cdot a]_{mn} = [m \cdot a']_{mn}$$

holds for all $a, a' \in \mathbf{Z}$. This is definitionally equivalent to

$$n|(a - a') \quad \Rightarrow \quad mn|(m \cdot a - m \cdot a'),$$

which is clear: if $a - a' = n \cdot d$, then

$$m \cdot a - m \cdot a' = m \cdot (a - a') = mn \cdot d.$$

Problem 3. Find an element $x \in \mathbf{Z}/23\mathbf{Z}$ satisfying

$$[16] \cdot x = [1].$$

Solution. We compute:

$$23 = 1 \cdot 16 + 7$$

$$16 = 2 \cdot 7 + 2$$

$$7 = 3 \cdot 2 + 1$$

so

$$1 = 7 - 3 \cdot 2$$

$$= 7 - 3 \cdot (16 - 2 \cdot 7)$$

$$= (-3) \cdot 16 + 7 \cdot 7$$

$$= (-3) \cdot 16 + 7 \cdot (24 - 1 \cdot 16)$$

$$= (-10) \cdot 16 + 7 \cdot 24$$

Therefore, $x = [-10]$.